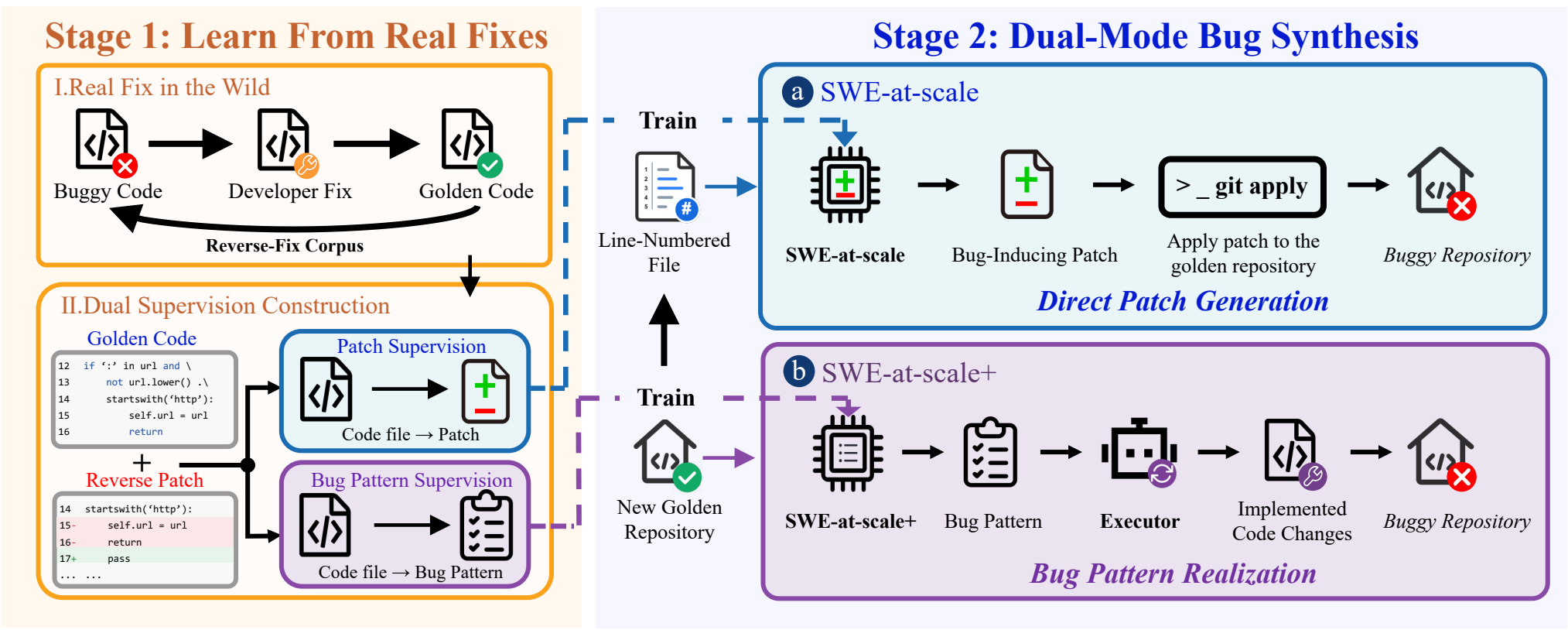

Stage 1: Learn From Real Fixes
I.Real Fix in the Wild
Buggy Code
Developer Fix
Golden Code
Reverse-Fix Corpus
II.Dual Supervision Construction
Golden Code
12 if ':' in url and \
13 not url.lower() .\
14 startswith('http'):
15 self.url = url
16 return
+
Reverse Patch
14 startswith('http'):
15- self.url = url
16- return
17+ pass
... ...
Patch Supervision
Code file → Patch
Bug Pattern Supervision
Code file → Bug Pattern
Train
Line-Numbered File
Train
New Golden Repository
Stage 2: Dual-Mode Bug Synthesis
a SWE-at-scale
SWE-at-scale
Bug-Inducing Patch
> _ git apply
Apply patch to the golden repository
Buggy Repository
Direct Patch Generation
b SWE-at-scale+
SWE-at-scale+
Bug Pattern
Executor
Implemented Code Changes
Buggy Repository
Bug Pattern Realization